DERIVED HOM AGAINST A NICE COMPLEX PUSHES FORWARD TO A PERFECT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete, being a direct reduction to the counted tensor-perfect lemma via the dual complex

Editorial note. The label lemma-ext-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- dual perfect complexes and the identification RSheafHom(E, G) = E^vee tensor G for perfect E
- the tensor-perfect direct image lemma (counted earlier in this lane)
- bounded complexes of coherent modules flat over the base with proper support

Source reference. Original reference 09J5: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5794–5794. Prerequisite pointer invoked at proof line(s) 5835; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JJ: sources/stacks-chow/tags.txt, source lines 10852–10852. Prerequisite pointer invoked at proof line(s) 5832; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let B be a Noetherian algebraic space over S. Let $f : X \to B$ be a morphism of algebraic spaces which is locally of finite type and quasi-separated. Let $E \in D(\mathcal{O}_X)$ be perfect. Let $\mathcal{G}^\bullet$ be a bounded complex of coherent $\mathcal{O}_X$-modules flat over B with support proper over B. Then $K = Rf_*R\mathcal{H}om(E, \mathcal{G})$ is a perfect object of $D(\mathcal{O}_B)$.*

Proof. Since E is a perfect complex there exists a dual perfect complex $E^\vee$, see Cohomology on Sites, Lemma 08JJ. Observe that $R\mathcal{H}om(E, \mathcal{G}^\bullet) = E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet$. Thus the perfectness of K follows from Lemma 09J5. □

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